Symmetric Matrices

Geometric Algorithms

CAS CS 132

Keywords

linear models

design matrices

general linear regression

symmetric matrices

the spectral theorem

orthogonal diagonalizability

quadratic forms

definiteness

constrained optimization

Recall: Symmetric Matrices

Def. A square matrix A is **symmetric** if $A^T = A$

Orthogonal Eigenvectors

Thm. For a symmetric matrix A, if $\mathbf{u}$ and $\mathbf{v}$ are eigenvectors for *distinct* eigenvalues, then $\mathbf{u}$ and $\mathbf{v}$ are orthogonal

Recall: Diagonalizable Matrices

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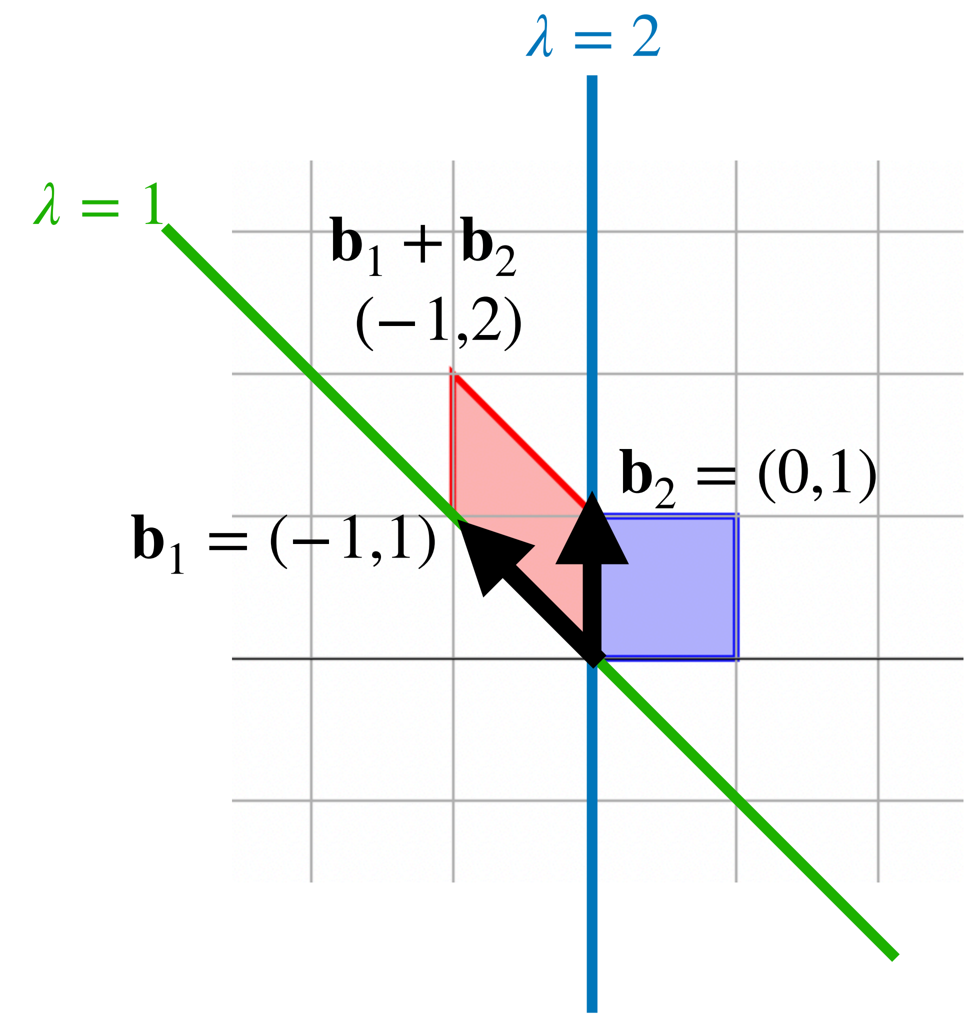
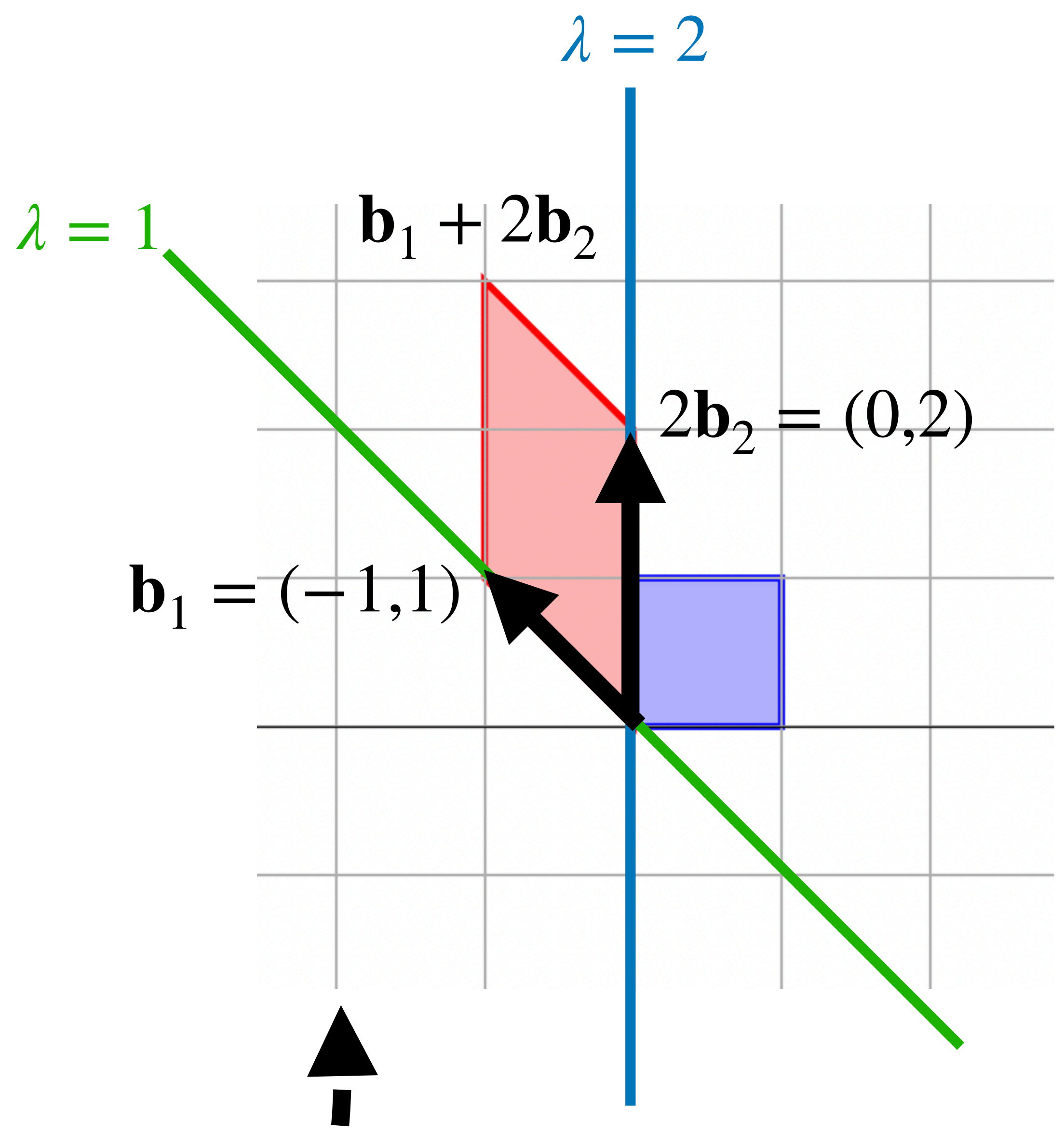
Diagonalizable matrices are the same as scaling matrices up to a change of basis

Recall: The Picture

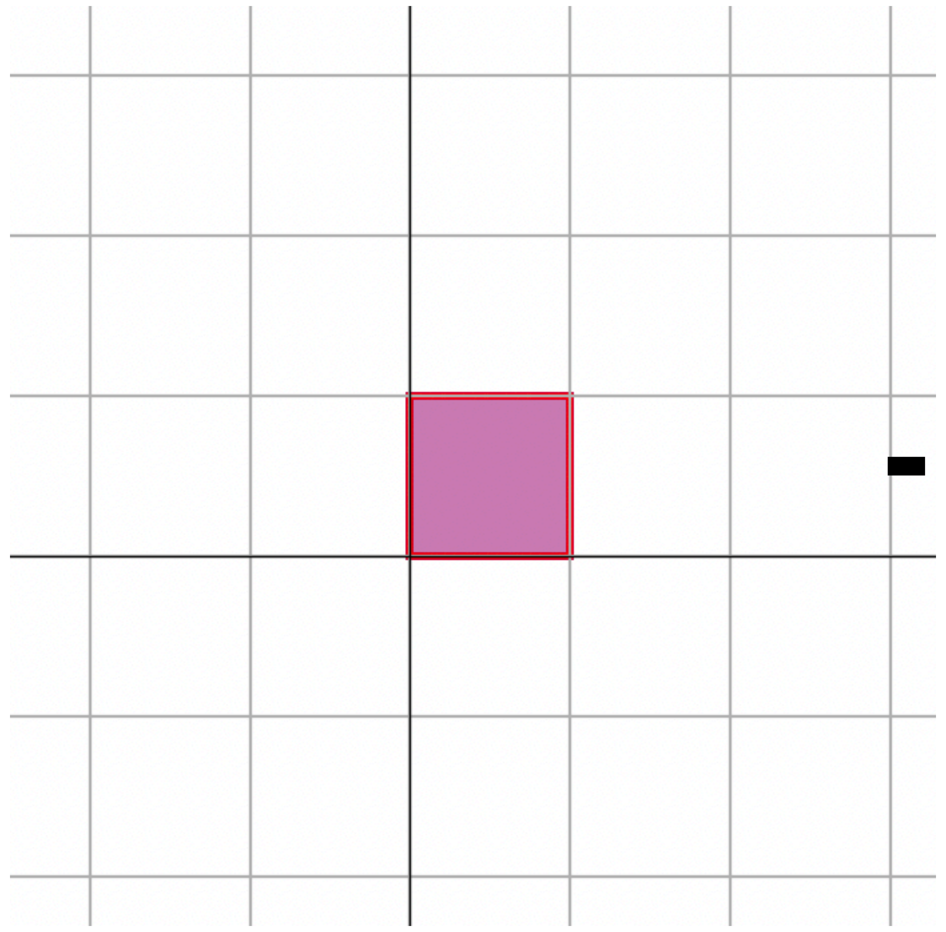


$$A = PDP^{-1}$$

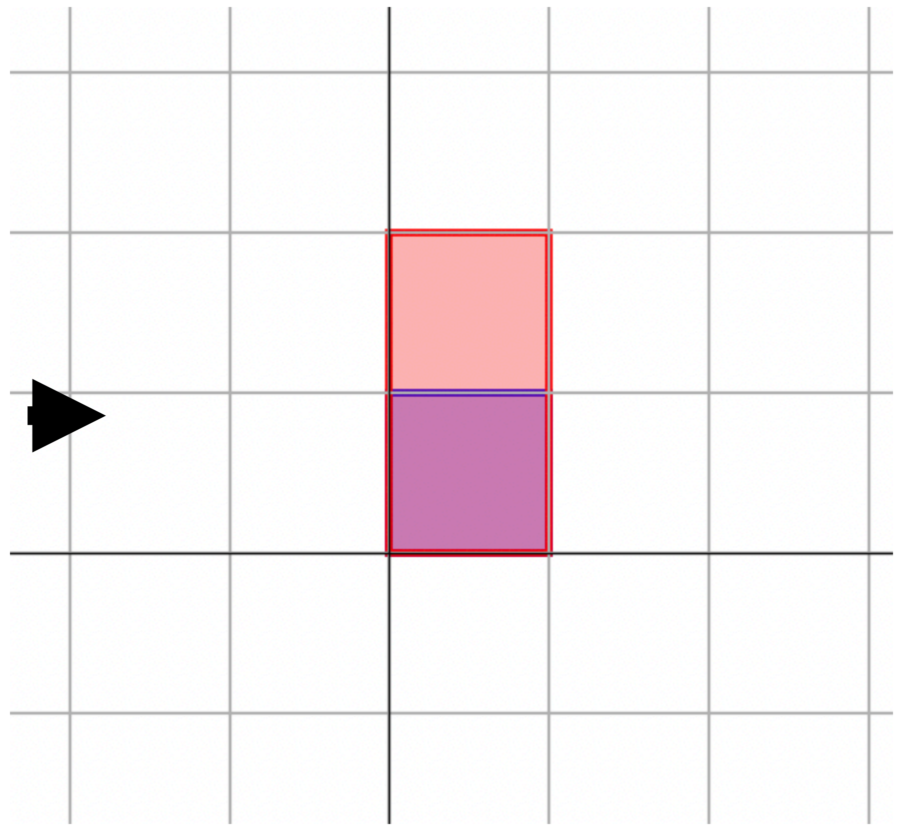
$$\begin{bmatrix} 2 & 0 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix}^{-1}$$



P^{-1}



D



P

Recall: The Diagonalization Theorem

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The idea:

Recall: The Diagonalization Theorem

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Thm. A is diagonalizable $\Leftrightarrow$ A has an eigenbasis

The idea:

» The columns of P form an **eigenbasis** for A

Recall: The Diagonalization Theorem

$$\underbrace{A = \overset{\text{eigenbasis}}{P}\underset{\text{eigenvalues}}{D}P^{-1}}$$

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The idea:

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» The diagonal of D are the eigenvalues for each column of P

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The idea:

» The columns of P form an **eigenbasis** for A

» The diagonal of D are the eigenvalues for each column of P

» **The matrix P^{-1} is a change of basis to this eigenbasis of A**

The Spectral Theorem

Thm. If A is symmetric then it has an ***orthonormal*** eigenbasis

Symmetric matrices are *diagonalizable*, and we can choose P to be ***orthonormal***

Recall: Orthonormal Matrices

Def. A matrix is **orthonormal** if its columns form an orthonormal set

This is sometimes called an **orthogonal** matrix

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And this is incredibly confusing, but we'll try to be consistent and clear

Recall: Inverses of Orthogonal Matrices

Thm. If a matrix $U \in \mathbb{R}^{n \times n}$ is orthogonalthen it is invertible and

$$U^{-1} = U^T$$

Orthogonal Diagonalizability

Def. A matrix A is **orthogonally diagonalizable** if there is a diagonal matrix D and matrix P such that

$$A = PDP^{-1} = PDP^{T}$$

P must be an *orthonormal matrix*

Symmetric matrices are orthogonally diagonalizable

Orthogonal Diagonalizability and Symmetry

Thm. $A \in \mathbb{R}^{n \times n}$ is orthogonally diagonalizable if and only if A is symmetric

You should know how to…

- Determine by looking at a matrix if it is orthogonally diagonalizable
- Determine an orthonormal eigenbasis of a (small) symmetric matrix by hand
- Find an orthogonal diagonalization of a (small) matrix by hand

Quadratic Forms

Quadratic Forms

Def. A **quadratic form** over the variables $x_1, \ldots, x_n$ is an multivariate polynomial $Q(x_1, \ldots, x_n)$ in which every term has degree two

Quadratic Forms and Symmetric Matrices

Fact. Quadratic forms can be represented as

$$Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$$

where A is *symmetric*

Example: Computing the Quadratic Form for a Matrix

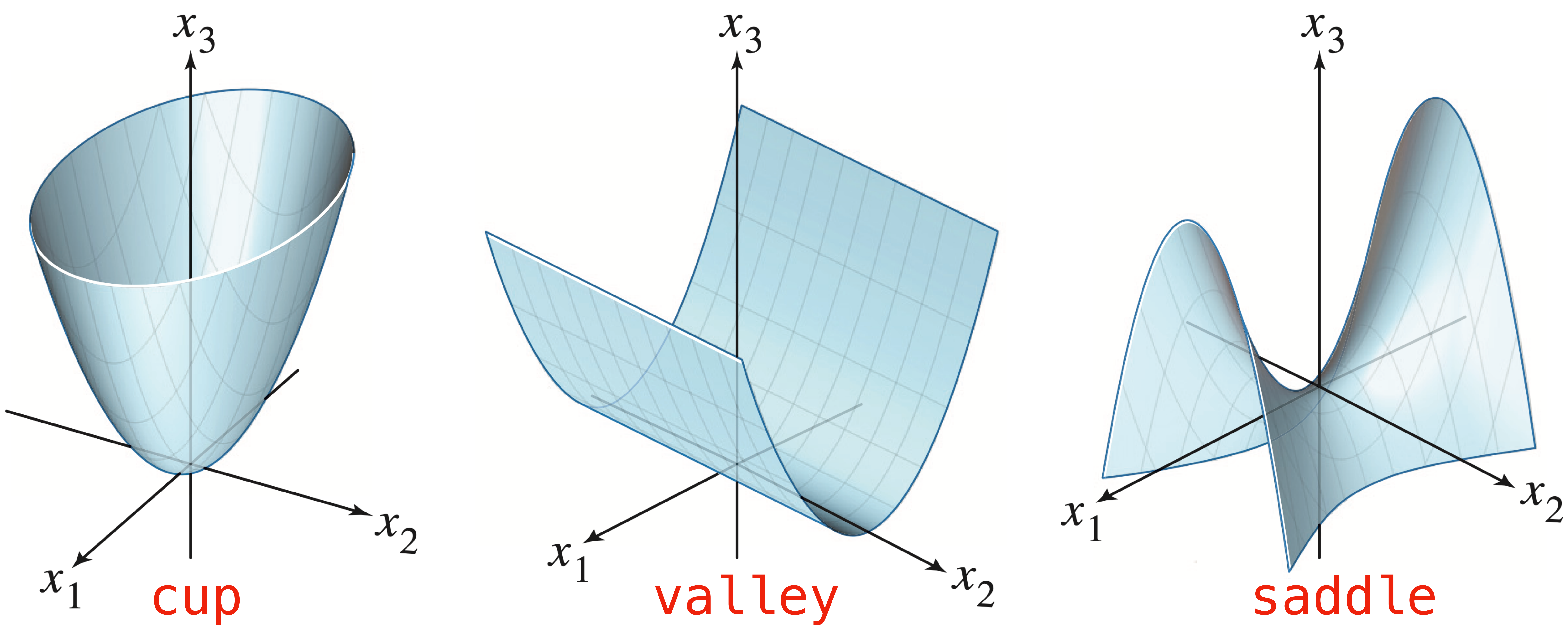
$$A = \begin{bmatrix} 3 & -2 \\ -2 & 7 \end{bmatrix}$$

This means, given a symmetric matrix A, we can compute its corresponding quadratic form

In General

$$\mathbf{x}^T A \mathbf{x} = \sum_{i=1}^{n} \sum_{j=1}^{n} A_{ij} x_i x_j = \sum_{i=1}^{n} A_{ii} x_i^2 + \sum_{i \neq j} (A_{ij} + A_{ji}) x_i x_j$$

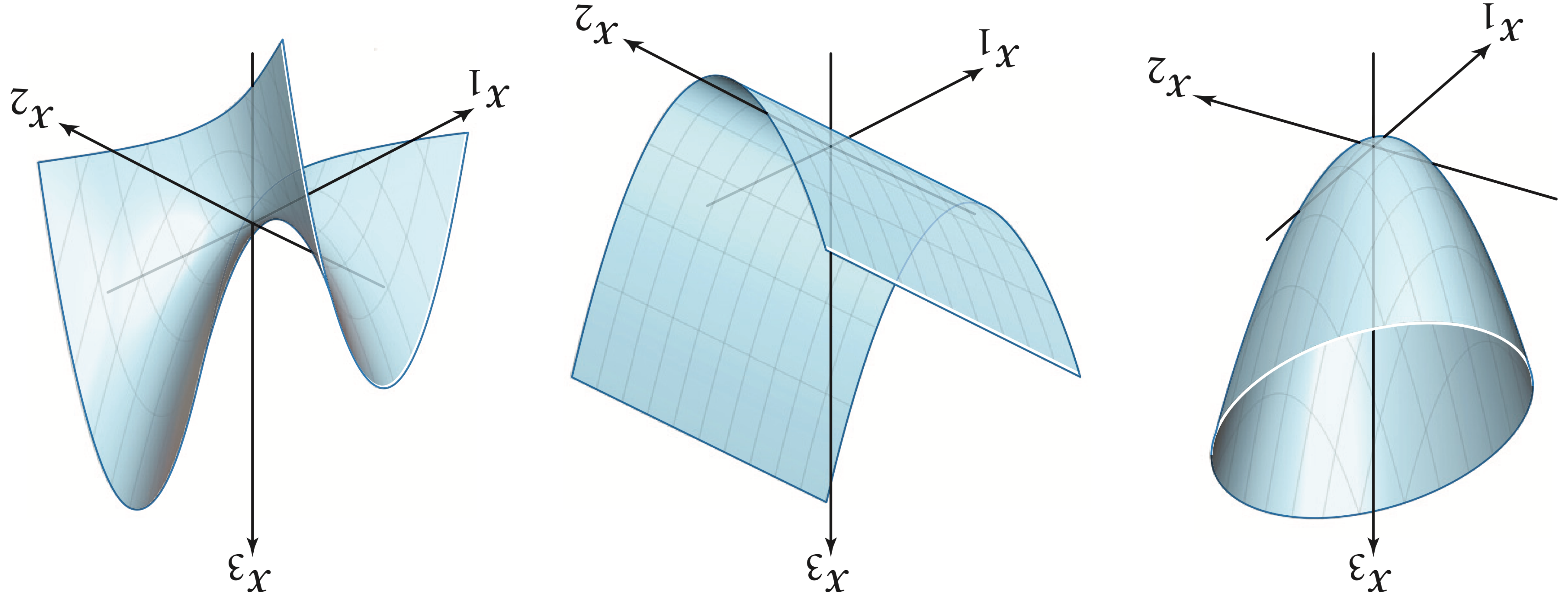
Shapes of of Quadratic Forms



There are three possible shapes (six if you include the negations)

Can we determine what shape it will be mathematically?

Shapes of of Quadratic Forms



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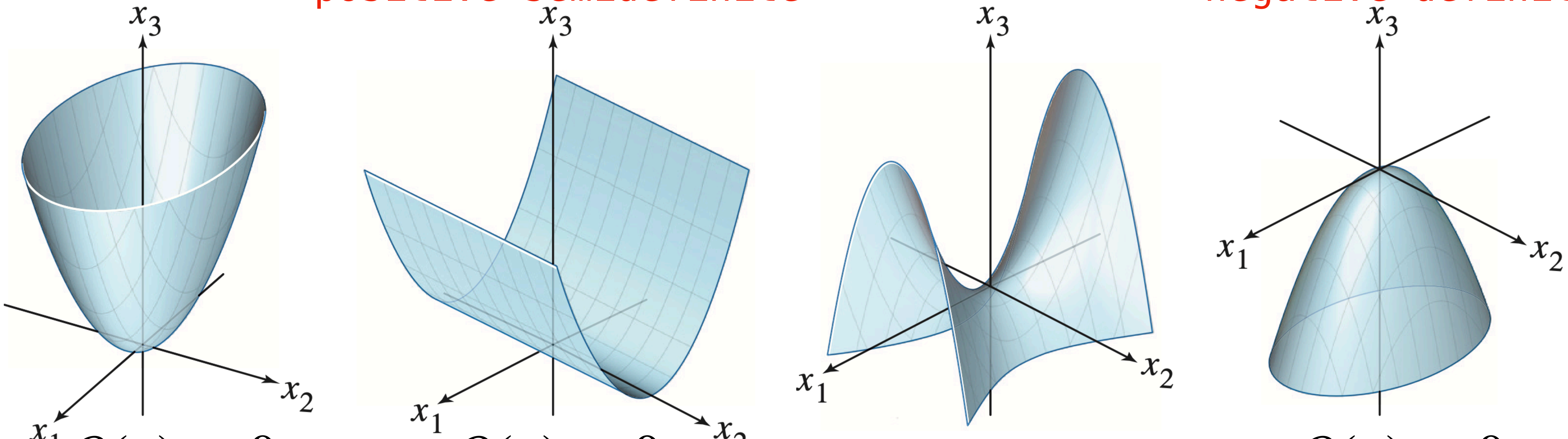
Linear Algebra and its Applications, Lay, Lay, McDonald

Definiteness

positive semidefinite

negative definite



$Q(\mathbf{x}) > 0$ $\quad Q(\mathbf{x}) \geq 0$ $\quad Q(\mathbf{x})$ can be + & – $\quad Q(\mathbf{x}) < 0$

positive definite

indefinite

For $\mathbf{x} \neq \mathbf{0}$ the graphs satisfy the given properties

Definiteness and Eigenvectors

Thm. For symmetric A the quad form $Q(\mathbf{x}) = \mathbf{x}^T A\mathbf{x}$ is

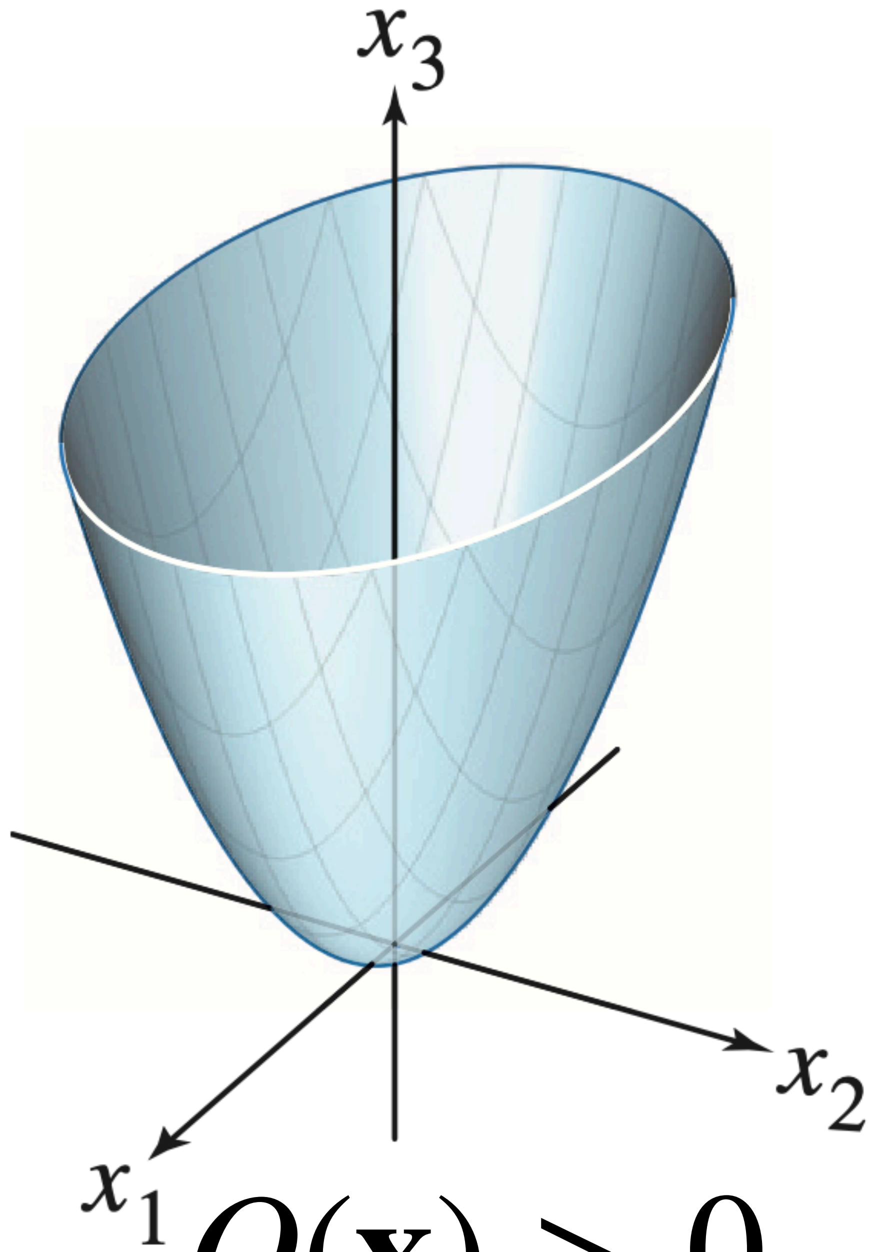
» **positive definite** ≡ all *positive* eigenvalues

» **positive semidefinite** ≡ all *nonnegative* eigenvalues

» **indefinite** ≡ positive and negative eigenvalues

» **negative definite** ≡ all *negative* eigenvalues

Definiteness



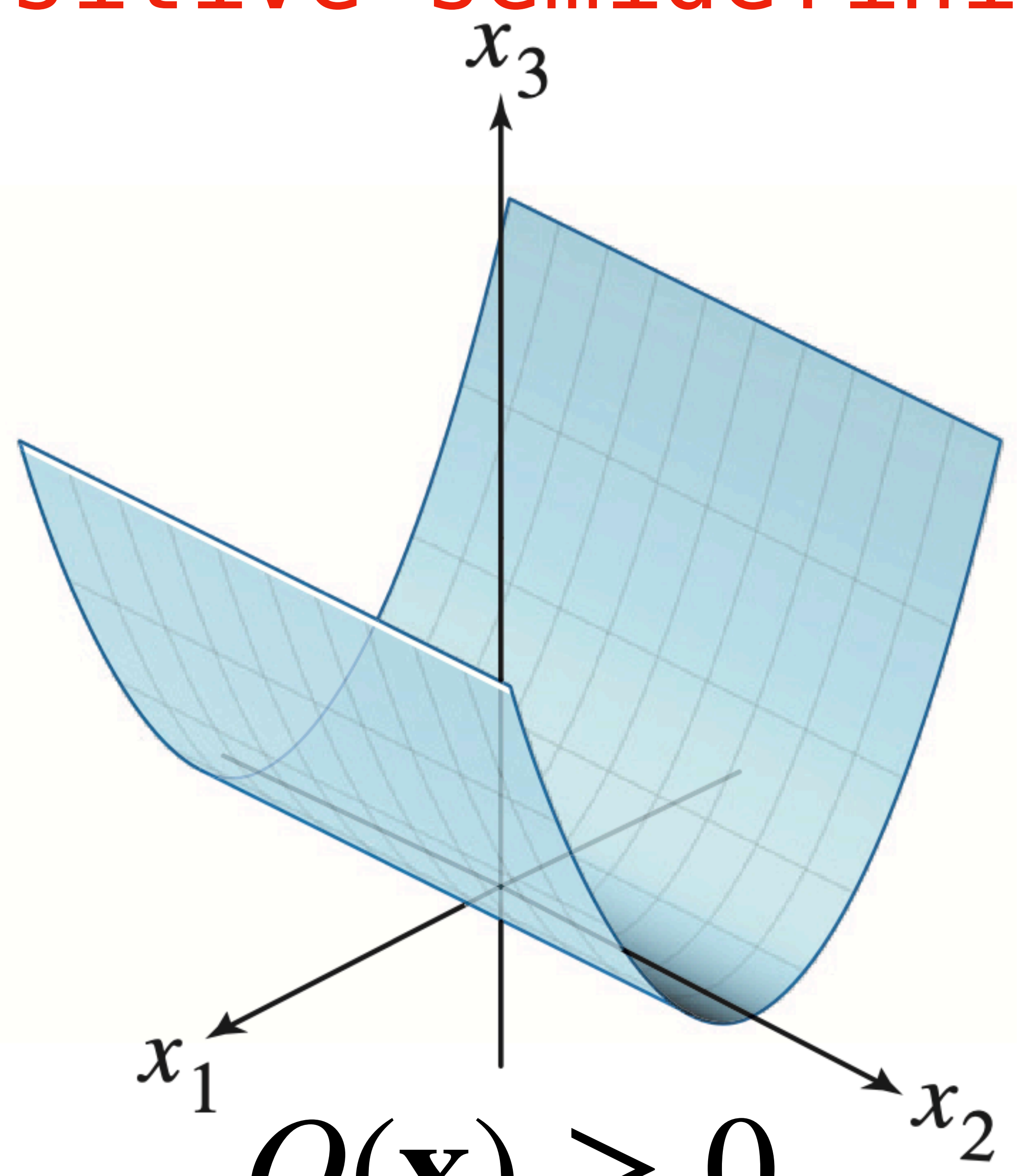
$Q(\mathbf{x}) > 0$

positive definite

all pos. eigenvals

all nonneg. eigenvals

positive semidefinite



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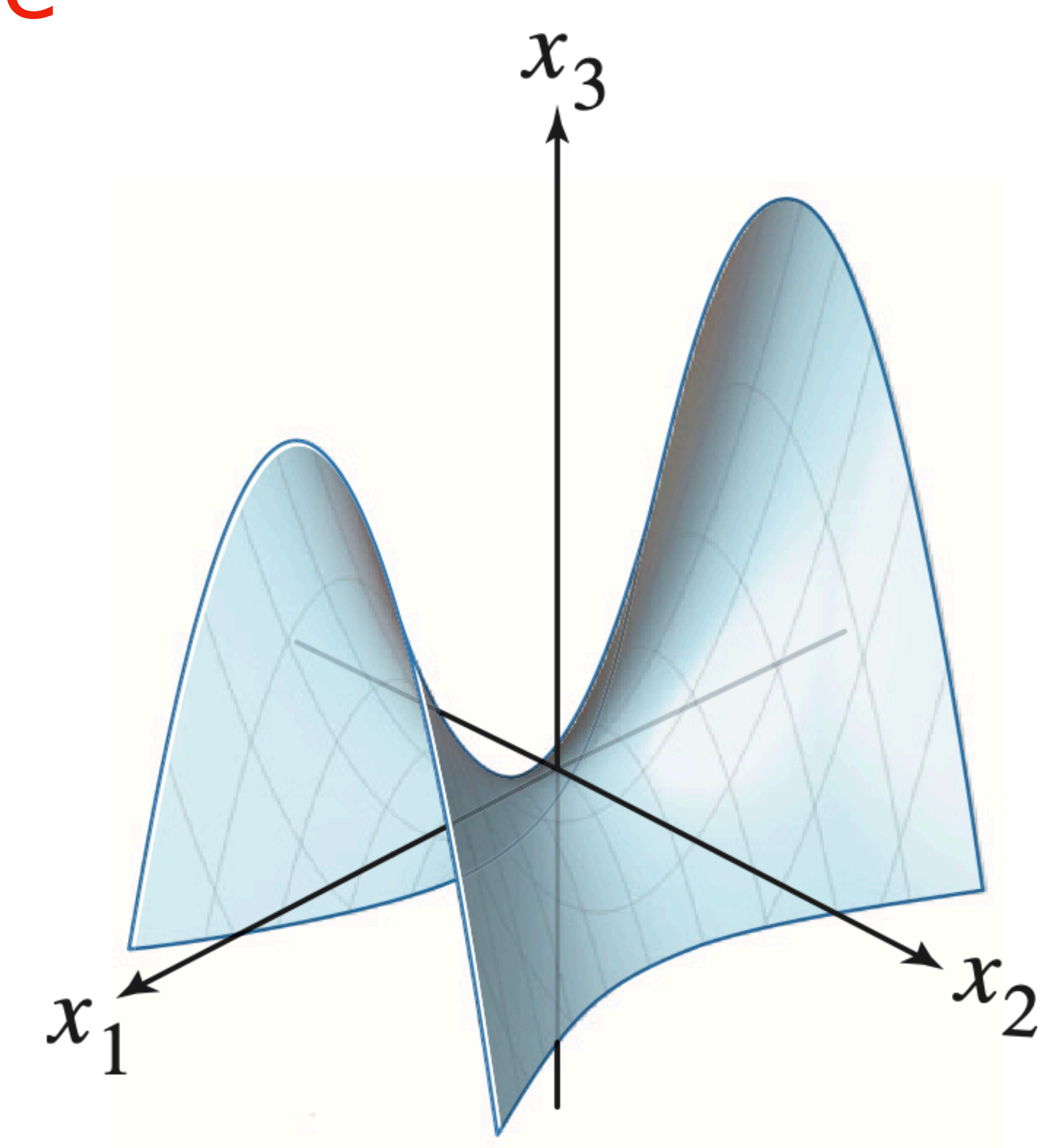


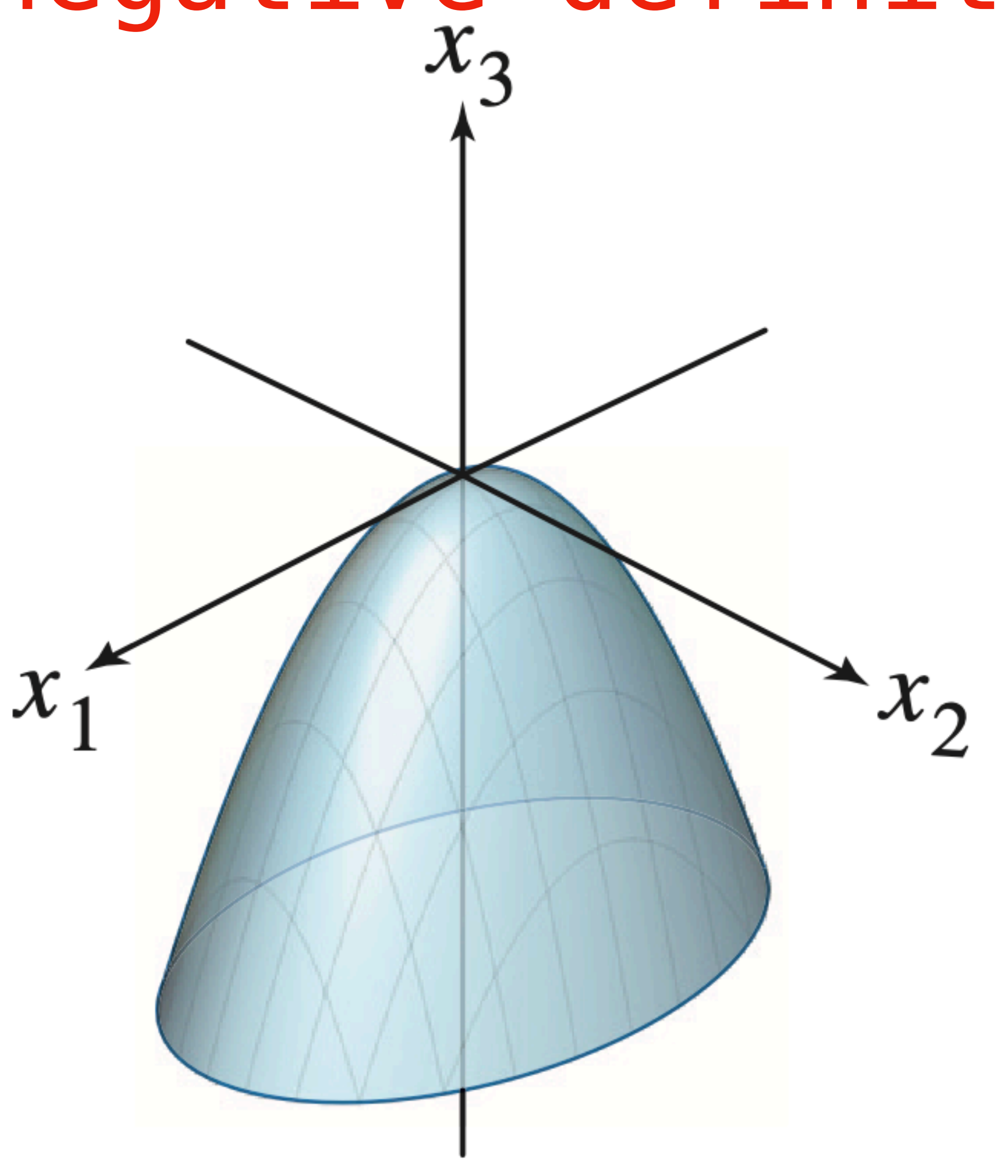
$Q(\mathbf{x})$ can be + & –

indefinite

pos. and neg. eigenvals

all neg. eigenvals

negative definite



$Q(\mathbf{x}) < 0$

You should know how to…

- Determine the underlying symmetric matrix of a quadratic form
- Expand a quadratic form written as $\mathbf{x}^T A\mathbf{x}$ into a multivariate polynomial
- Determine the definiteness of the quadratic form/symmetric matrix

Constrained Optimization

In General

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Given a function $f: \mathbb{R}^n \to \mathbb{R}$ and a set of vectors X from $\mathbb{R}^n$ the **constrained minimization problem** for f over X is the problem of determining

$$\min_{\mathbf{v} \in X} f(\mathbf{v}) \quad \text{or} \quad \max_{\mathbf{v} \in X} f(\mathbf{v})$$

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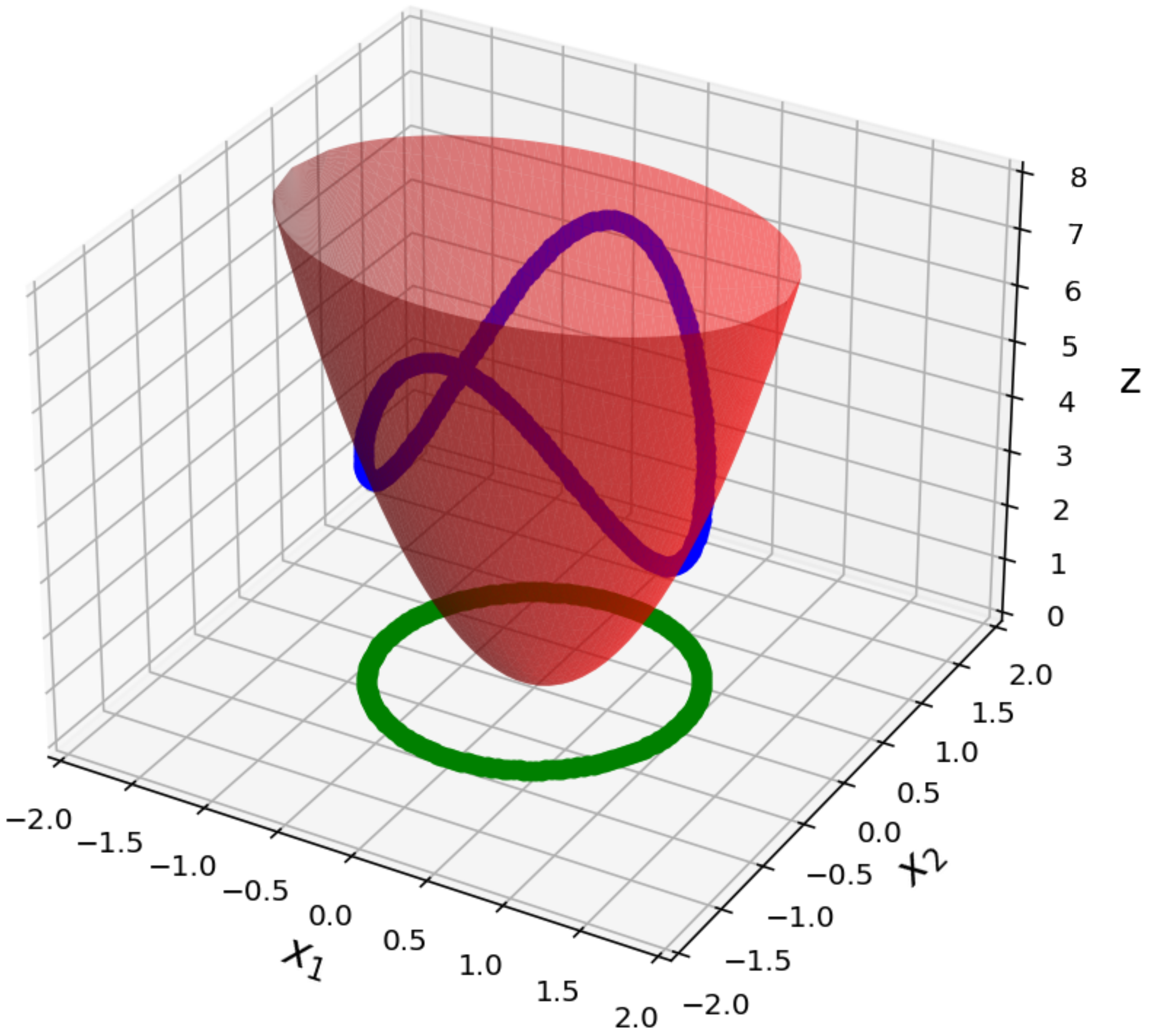
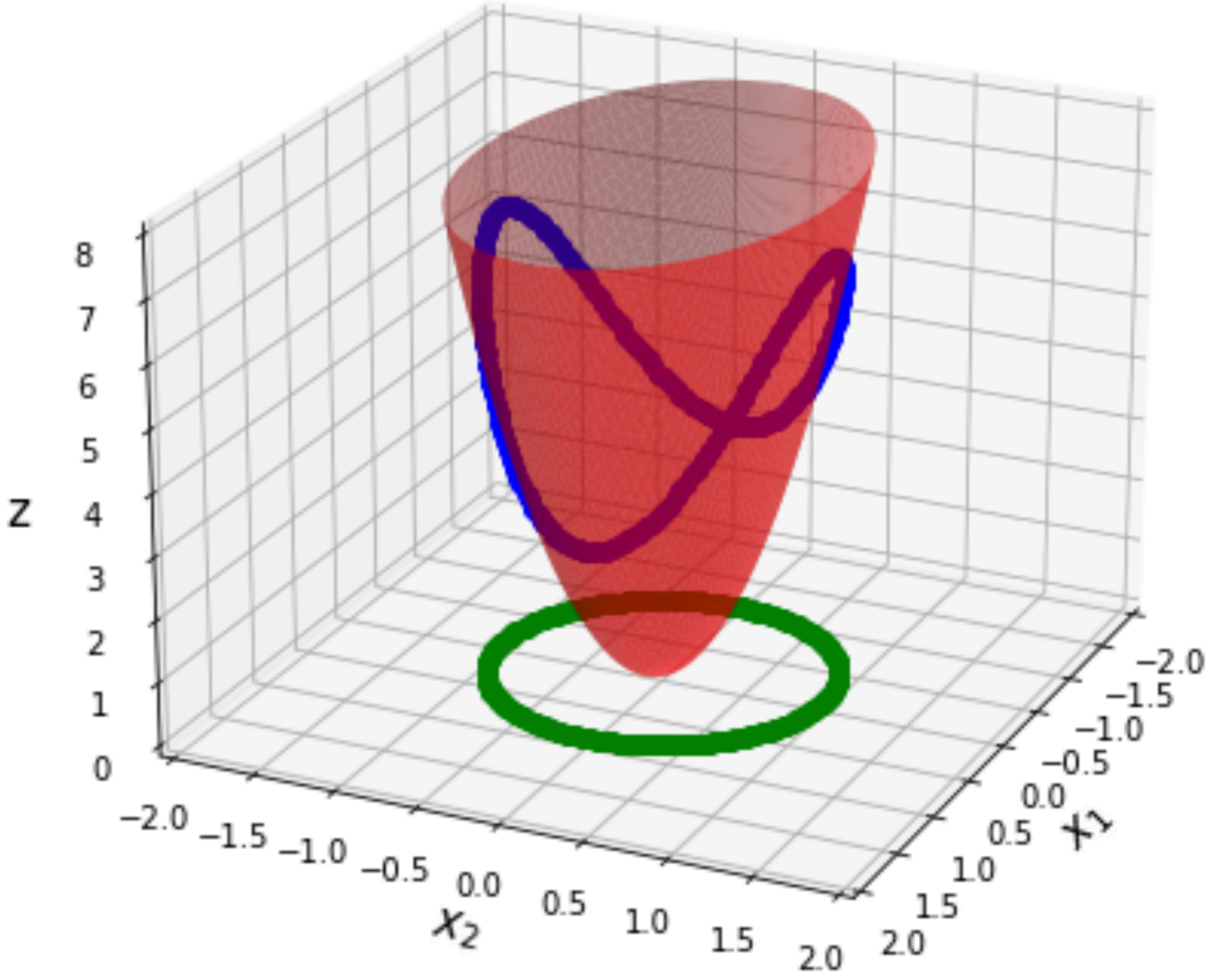
Determine the smallest value of $f(\mathbf{v})$ ***subject to*** *choosing a vector in X*

Constrained Optimization for Quadratic Forms and Unit Vectors

mini/maximize $\mathbf{x}^T A\mathbf{x}$ subject to $\|\mathbf{x}\| = 1$



Example: $3x_1^2 + 7x_2^2$



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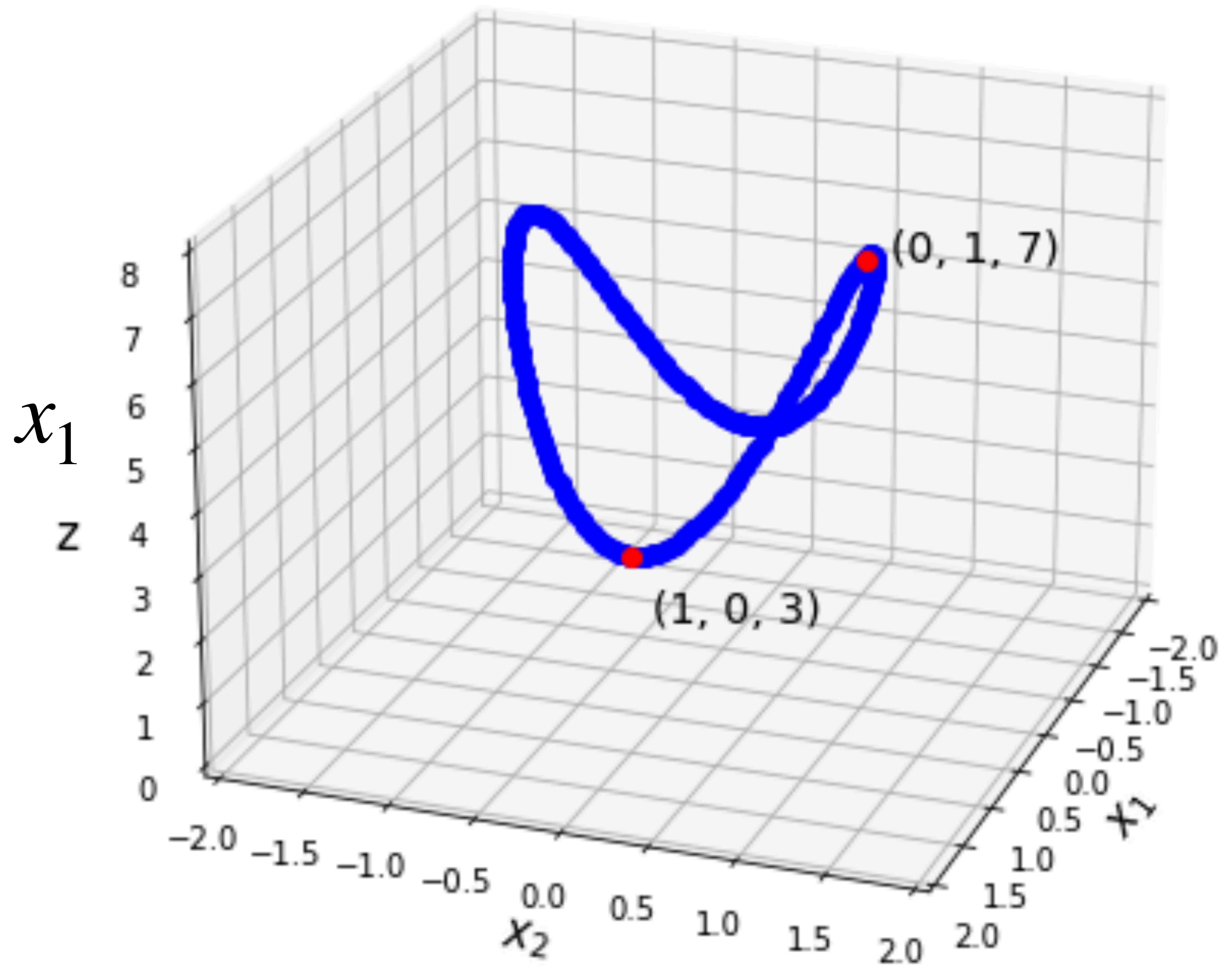

Constrained Optimization and Eigenvalues

Thm. For a symmetric matrix A with *largest* eigenvalue λ_1 and *smallest* eigenvalue λ_n

$$\max_{\|\mathbf{x}\|=1} \mathbf{x}^T A\mathbf{x} = \lambda_1 \qquad \min_{\|\mathbf{x}\|=1} \mathbf{x}^T A\mathbf{x} = \lambda_n$$

No matter what A looks like, this holds

You should know how to...

- Determine $\max_{\|\mathbf{x}\|=1} \mathbf{x}^T A\mathbf{x}$ and $\min_{\|\mathbf{x}\|=1} \mathbf{x}^T A\mathbf{x}$ for a given symmetric matrix A
- Determine $\max_{\|\mathbf{x}\|=1} Q(\mathbf{x})$ and $\min_{\|\mathbf{x}\|=1} Q(\mathbf{x})$ for a given quadratic form $Q(\mathbf{x})$

workshop